Subject : Burgers equation (**Preliminary results**)
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Date : 2026-06-06 17:45:48

1 Burgers equation

The Burgers equation reads (Debnath (2008)):

$$\boxed{\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} - \nu\frac{\partial^2 u}{\partial x^2} = 0, \quad u > 0} \tag{1}$$

The Burgers equation is written in conservative form, because we will use a finite volume method to solve this equation, it reads:

$$\frac{\partial u}{\partial t} + \frac{1}{2}\frac{\partial u^2}{\partial x} - \frac{\partial}{\partial x}\left(\nu\frac{\partial u}{\partial x}\right) = 0 \tag{2}$$

Applying the finite volume approach:

$$\int_\Omega \frac{\partial u}{\partial t}\, d\Omega + \frac{1}{2}\int_\Omega \frac{\partial u^2}{\partial x}\, d\Omega \; - \int_\Omega \frac{\partial}{\partial x}\left(\nu\frac{\partial u}{\partial x}\right)\, d\Omega = 0 \tag{3}$$

and after applying Green's theorem it yields:

$$\int_\Omega \frac{\partial u}{\partial t}\, d\Omega + \frac{1}{2}\oint_\Gamma u^2\, d\Gamma - \oint_\Gamma \nu\frac{\partial u}{\partial x}\, d\Gamma = 0 \tag{4}$$

2 Discretization in time and space

The discretization in space and the discretization in space are discussed in separate sections. Due to the applying regularization procedure the viscosity (ν) is increased by an artificial viscosity (Ψ), so the the implementation of $\nu + \Psi$ is a function of x.

2.1 Discretization in time

The used discretization in time reads:

$$\begin{aligned}\int_\Omega \frac{\partial u}{\partial t}\, d\Omega \approx \Delta x\Delta t_{inv}\left(\frac{1}{8}\Delta u_{i-1}^{n+1,p+1} + \frac{6}{8}\Delta u_i^{n+1,p+1} + \frac{1}{8}\Delta u_{i+1}^{n+1,p+1}\right) + \\ + \Delta x\Delta t_{inv}\left(\frac{1}{8}\left(u_{i-1}^{n+1,p} - u_{i-1}^{n}\right) + \frac{6}{8}\left(u_i^{n+1,p} - u_i^{n}\right) + \frac{1}{8}\left(u_{i+1}^{n+1,p} - u_{i+1}^{n}\right)\right)\end{aligned} \tag{5}$$

2.2 Discretization in place

Convection-term

$$\frac{1}{2}\oint_\Gamma u^2\, d\Gamma = \frac{1}{2}\left. u^2\right|_{i+\frac{1}{2}} - \frac{1}{2}\left. u^2\right|_{i-\frac{1}{2}} \tag{6}$$

with the approximation $u^2 = (u^{n+\theta,p})^2 + 2u^{n+\theta,p}\theta\Delta u$ the discretization reads:

$$\approx u^{n+\theta,p}_{i+\frac{1}{2}}\theta\Delta u_{i+\frac{1}{2}} - u^{n+\theta,p}_{i-\frac{1}{2}}\theta\Delta u_{i-\frac{1}{2}} + \frac{1}{2}\left(u^{n+\theta,p}_{i+\frac{1}{2}}\right)^2 - \frac{1}{2}\left(u^{n+\theta,p}_{i-\frac{1}{2}}\right)^2 \tag{7}$$

Viscosity term

With $\varepsilon = \nu + \Psi$ the viscosity term reads:

$$-\oint_\Gamma \varepsilon\frac{\partial u}{\partial x}\, d\Gamma = -\left(\left(\varepsilon\frac{\partial u}{\partial x}\right)_{i+\frac{1}{2}} - \left(\varepsilon\frac{\partial u}{\partial x}\right)_{i-\frac{1}{2}}\right) \tag{8}$$

approximated by

$$\Rightarrow -\theta\frac{\varepsilon_{i+\frac{1}{2}}}{\Delta x}\Delta u^{n+1,p+1}_{i+1} + \theta\left(\frac{\varepsilon_{i+\frac{1}{2}}}{\Delta x} + \frac{\varepsilon_{i-\frac{1}{2}}}{\Delta x}\right)\Delta u^{n+1,p+1}_{i} - \theta\frac{\varepsilon_{i-\frac{1}{2}}}{\Delta x}\Delta u^{n+1,p+1}_{i-1} + \tag{9}$$

$$-\left\{\left(\varepsilon_{i+\frac{1}{2}}\frac{u^{n+\theta,p}_{i+1} - u^{n+\theta,p}_{i}}{\Delta x} - \varepsilon_{i-\frac{1}{2}}\frac{u^{n+\theta,p}_{i} - c^{n+\theta,p}_{i-1}}{\Delta x}\right)\right\} \tag{10}$$

3 Energy Equation

The **inviscid** Burgers' equation is given by:

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = 0. \tag{11}$$

The kinetic energy per unit mass is:

$$E = \frac{1}{2}u^2 \tag{12}$$

The time derivative of the energy reads:

$$\frac{\partial E}{\partial t} = \frac{\partial}{\partial t}\left(\frac{1}{2}u^2\right) = u\frac{\partial u}{\partial t} = u\left(-u\frac{\partial u}{\partial x}\right) = -u^2\frac{\partial u}{\partial x} = -\frac{\partial}{\partial x}\left(\frac{1}{3}u^3\right) \tag{13}$$

So the energy equation becomes:

$$\frac{\partial E}{\partial t} + \frac{\partial}{\partial x}\left(\frac{1}{3}u^3\right) = 0 \tag{14}$$

4 Boundary

Because it is a one dimensional equation we have only boundary conditions at the west- and east-side of the domain. We consider only a flow direction in positive x-direction, so $u > 0$.

4.1 West boundary

This is an inflow boundary ($u > 0$).

$$u^{n+1} = f(t) \tag{15}$$

$$u^{n+\theta,p} + \theta\Delta u = f(t) \tag{16}$$

$$\theta\Delta u = f(t) - u^{n+\theta,p} \tag{17}$$

4.2 East boundary

This is an outflow boundary ($u > 0$).

Not yet documented

5 Energy Equation of inviscid Burgers equation

The **inviscid Burgers equation** is given by:

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = 0 \tag{18}$$

where $u = u(x,t)$ is the velocity field.

The kinetic energy per unit mass is:

$$E = \frac{1}{2}u^2 \tag{19}$$

We want an equation governing E. Differentiate E with respect to time

$$\frac{\partial E}{\partial t} = \frac{\partial}{\partial t}\left(\frac{1}{2}u^2\right) = u\frac{\partial u}{\partial t} \tag{20}$$

$$\frac{\partial E}{\partial t} = u\left(-u\frac{\partial u}{\partial x}\right) = -u^2\frac{\partial u}{\partial x} \tag{21}$$

So the energy equation becomes:

$$\frac{\partial E}{\partial t} + \frac{\partial}{\partial x}\left(\frac{1}{3}u^3\right) = 0 \tag{22}$$

This shows that energy is conserved in the inviscid Burgers' equation .

6 Numerical experiment

In this section we will show the results of the Burgers equation with a Dirichlet boundary conditions at the left side of the domain.

The considered Burgers equation reads:

$$\frac{\partial u}{\partial t} + \frac{1}{2}\frac{\partial u^2}{\partial x} - \frac{\partial}{\partial x}\left((\nu + \Psi)\frac{\partial u}{\partial x}\right) = 0, \qquad u > 0. \tag{23}$$

where

u	Convected velocity, $[\mathrm{m\,s^{-1}}]$.
ν	Diffusion coefficient, $[\mathrm{m^2\,s^{-1}}]$.
Ψ	Artificial diffusion coefficient, $[\mathrm{m^2\,s^{-1}}]$.

The prescribed boundary condition for the velocity reads

$$u(0,t) = u_{given}(t) \begin{cases} \frac{1}{2}\left(\cos\left(\pi\frac{t_{reg}-t}{t_{reg}}\right)+1\right) & \text{if } t < t_{reg}, \\ 1 & \text{if } t \geq t_{reg}, \end{cases} \tag{24}$$

where

t_{reg}	The regularization time for the given time-series, [s]
$u_{given}(t)$	Prescribed (given) time-series at the boundary.

7 Experiments without/with artifcial viscosity

In this section results of simulations without and with artificial viscosity will be compared. Several test cases where the viscosity is well resolved on the grid and several cases where is not.

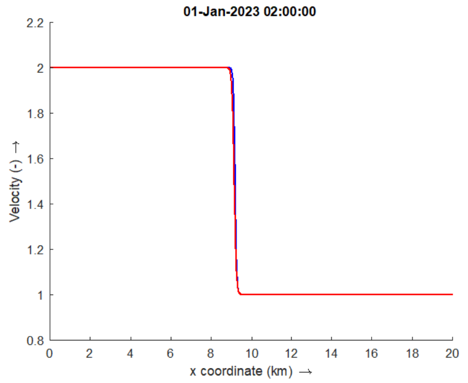
Numerical experiment

The numerical experiment is performed with the following parameters:

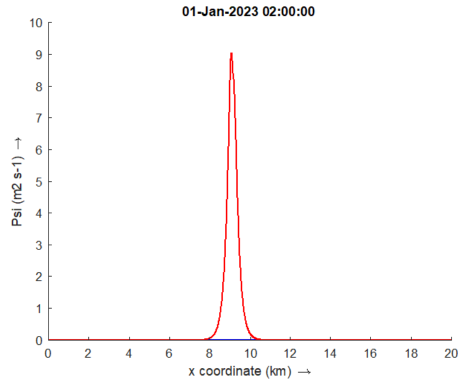
- ▷ Length of the domain, $L_x = 20\,000$ [m]
- ▷ Grid size, $\Delta x = 10$ [m]
- ▷ Start time, $t_{start} = 0$ [s]
- ▷ End time, $t_{stop} = 4$ [h]
- ▷ Timestep, $\Delta t = 5$ [s]
- ▷ Regularization time for time-series, $t_{reg} = 1800$ [s]
- ▷ Prescribed initial velocity, $u_{initial}(x,t) = 1$ $[\mathrm{m\,s^{-1}}]$
- ▷ Prescribed velocity on boundary at $x = 0$ is given by equation (24), with $u_{given} = 2$ $[\mathrm{m\,s^{-1}}]$.
- ▷ No boundary is prescribed at $x = 20\,000$ [m].

The speed of the fully developed shock wave for the inviscid Burgers equation is $u_{shock} = 1.5$ $[\mathrm{m\,s^{-1}}]$ (Debnath, 2008, eq. 8.1.67)

8 Experiment with viscosity of $20\ [\mathrm{m\,s^{-2}}]$

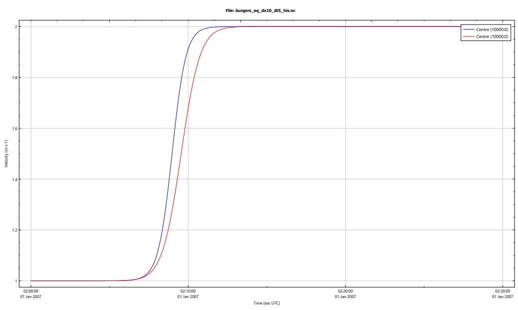


(a) *Velocity along channel at* $2\ [\mathrm{h}]$. *The blue line, without artificial viscosity and the red line with artificial velocity.*

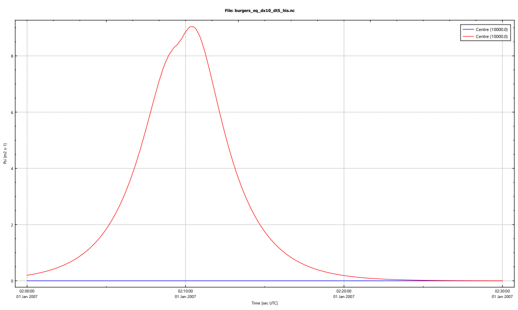


(b) *Used artificial viscoity* Ψ *along channel at* $2\ [\mathrm{h}]$. *The blue line, without artificial viscosity* ($\equiv 0$) *and the red line with artificial velocity.*

Figure 1: *Transect for velocity* $\nu = 20\ [\mathrm{m^2\,s^{-1}}]$

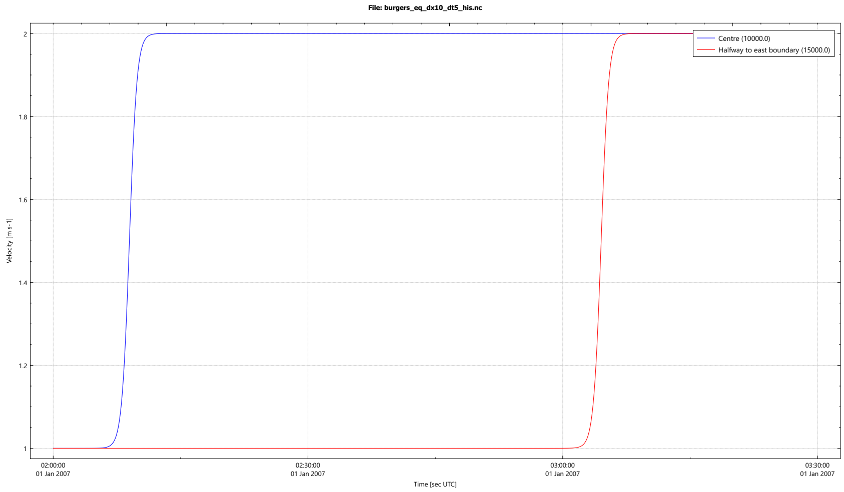


(a) *Time histories of the velocity for the observation point at the centre of the channel. The blue line, without artificial viscosity and the red line with artificial velocity.*

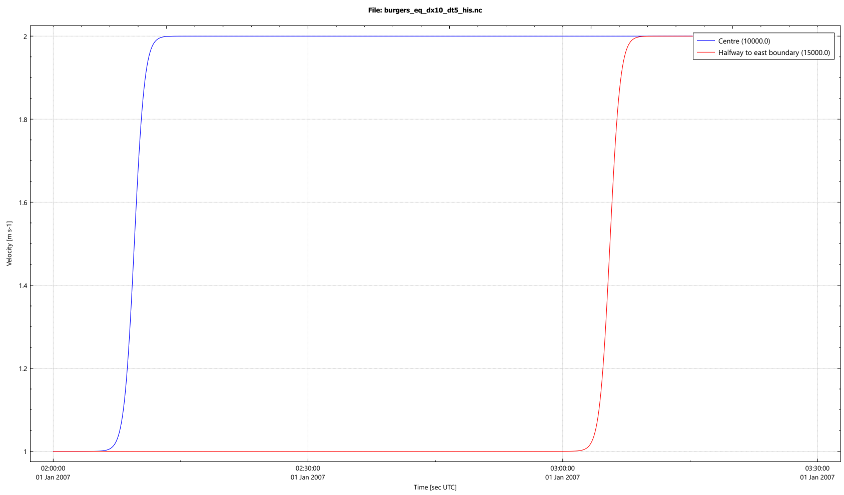


(b) *Time histories of the artificial viscosity* Ψ *for the observation point at the* $x = 10\,000\ [\mathrm{m}]$ *(blue line) and at* $x = 15\,000\ [\mathrm{m}]$ *(red line).*

Figure 2: *Time histories for* $\nu = 20\ [\mathrm{m^2\,s^{-1}}]$



(a) *Time histories of the velocity for the observation point at the* $x = 10\,000$ [m] *(blue line) and at* $x = 15\,000$ [m] *(red line).*

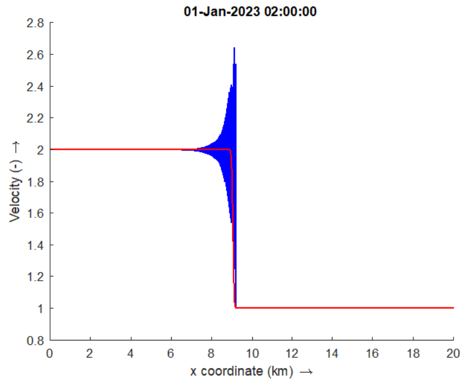


(b) *Time histories of the artificial viscosity* Ψ *for the observation point at the* $x = 10\,000$ [m] *(blue line) and at* $x = 15\,000$ [m] *(red line).*

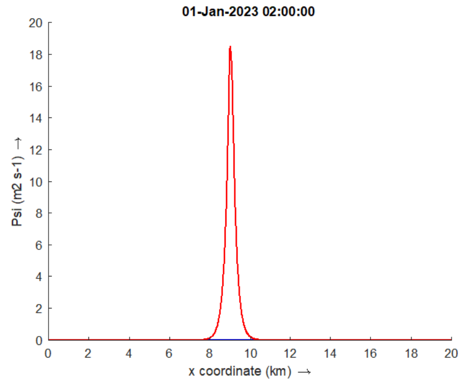
Figure 3: *Time histories for when* $\nu = 20$ $[\mathrm{m^2\,s^{-1}}]$

The wave front has a speed of $u_{shock} = 1.5$ $[\mathrm{m\,s^{-1}}]$ for both cases, without and with artificial viscosity (i.e. 1.5000 vs 1.4999 $[\mathrm{m\,s^{-1}}]$).

9 Experiment with viscosity of $0.1\ [\mathrm{m\,s^{-2}}]$

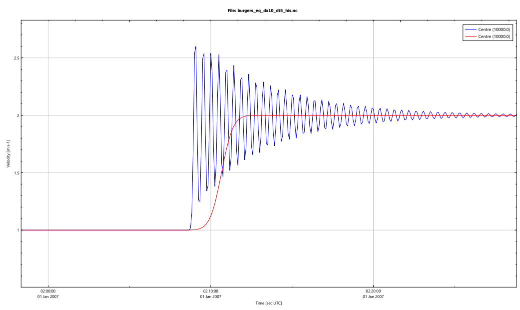


(a) *Velocity along channel at* $2\ [\mathrm{h}]$*. The blue line, without artificial viscosity and the red line with artificial velocity.*

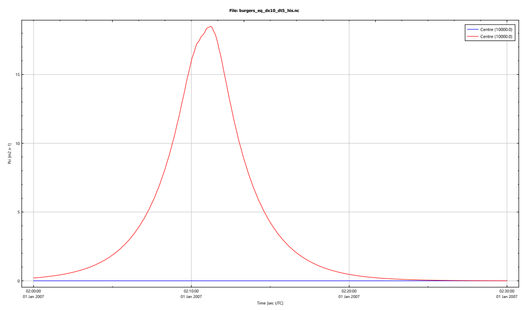


(b) *Used artificial viscoity* Ψ *along channel at* $2\ [\mathrm{h}]$*. The blue line, without artificial viscosity* $(\equiv 0)$ *and the red line with artificial velocity.*

Figure 4: *Transect for velocity* $\nu = 0.1\ [\mathrm{m^2\,s^{-1}}]$

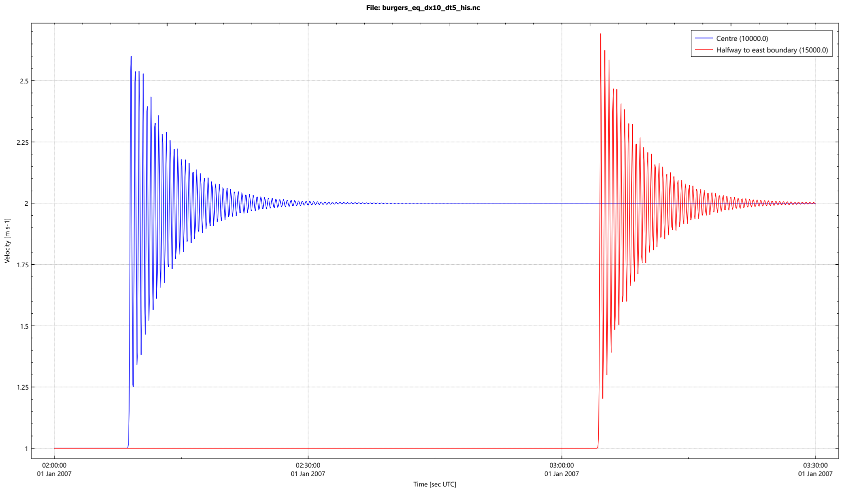


(a) *Time histories of the velocity for the observation point at the centre of the channel. The blue line, without artificial viscosity and the red line with artificial velocity.*

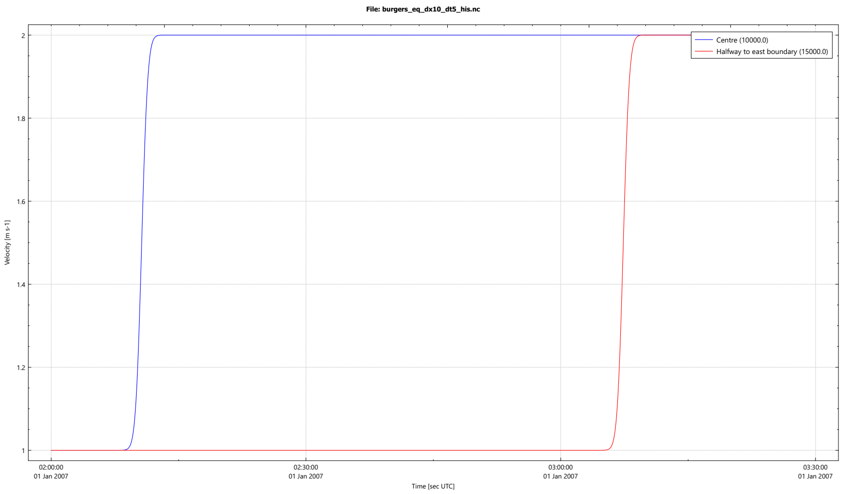


(b) *Time histories of the artificial viscosity* Ψ *for the observation point at the centre of the channel. The blue line, without artificial viscosity* $(\equiv 0)$ *and the red line with artificial velocity.*

Figure 5: *Time histories for* $\nu = 0.1\ [\mathrm{m^2\,s^{-1}}]$



***(a)** Time histories of the velocity for the observation point at the* $x = 10\,000$ [m] *(blue line) and at* $x = 15\,000$ [m] *(red line).*



***(b)** Time histories of the artificial viscosity* Ψ *for the observation point at the* $x = 10\,000$ [m] *(blue line) and at* $x = 15\,000$ [m] *(red line).*

***Figure 6:** Time histories for when* $\nu = 0.1\ [\mathrm{m}^2\,\mathrm{s}^{-1}]$

The wave front has a speed of $u_{shock} = 1.5\ [\mathrm{m\,s}^{-1}]$ for both cases, without and with artificial viscosity (i.e. 1.5000 vs $1.5000\ [\mathrm{m\,s}^{-1}]$).

10 Experiment with non linear source term

This experiment is taken from Colombo et al. (2026) and read

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} - \nu\frac{\partial^2 u}{\partial x^2} = u^2, \quad u > 0 \tag{25}$$

With viscosity coefficient $\nu = 0.0\ [\mathrm{m}^2\,\mathrm{s}^{-1}]$ and initial condition

$$u(x,0) = \exp(x) + 0.3\exp(-200(x+0.5)^2), \quad x \in= [-1,1] \tag{26}$$

and boundary condition at the left side

$$u(-1, t) = \exp(-1). \tag{27}$$

Not yet documented

References

Colombo, C., C. Dalmaso, L. O. Müller, and A. Siviglia (2026). “Well-balanced high-order method for non-conservative hyperbolic PDEs with source terms: Application to one-dimensional blood flow equations with gravity”. In: *Journal of Computation Physics*. File: Colombo_Application to one-dimensional blood_2026.pdf.

Debnath, Lokenath (2008). *Sir Lighthill and Modern Fluid Dynamics*. Imperial College Press.